

The slide features a dark red background with a decorative border. On the left side, there is a vertical strip of colorful books in red, green, blue, and purple. On the right side, there are two lanterns with yellow flames, one at the top and one at the bottom. The title 'Library Management System in C' is written in a large, white, serif font, centered on the slide.

Library Management System in C

Presented By- Tanushree Bali (1262251830)
Pranav Sahoo (1262250194)
Rohit Nukala (1262251929)
Pranav Patil (1262252853)
Mihir Gondse (1262252478)
Antonio Mauro (1262254184)

Objectives



01

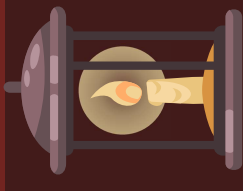
Add
Books

02

Display
Books

03

Search
Books



04

Issue
Books

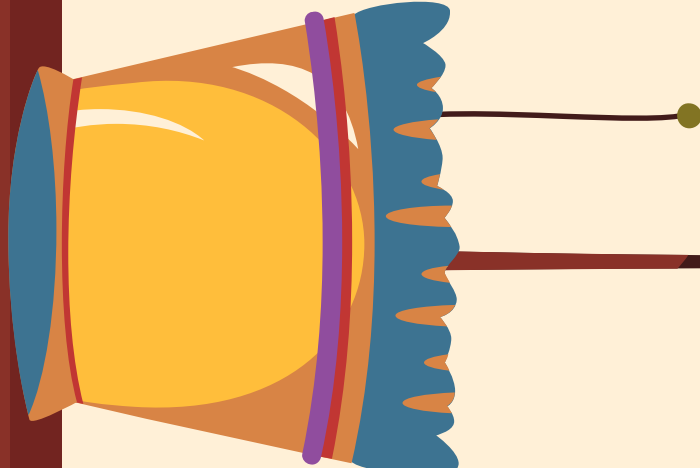
05

Return
Books

06

Exit
System





Arrays Used in the Program



Variable

Book_id[10]

Book_name[10][50]

Book_quantity[10]

Count

Purpose

Stores IDs of books

Stores names of books

Stores quantity available

Tracks number of books



Program Flow (Working)

Main Loop

while(1)

Program runs continuously until user chooses Exit.

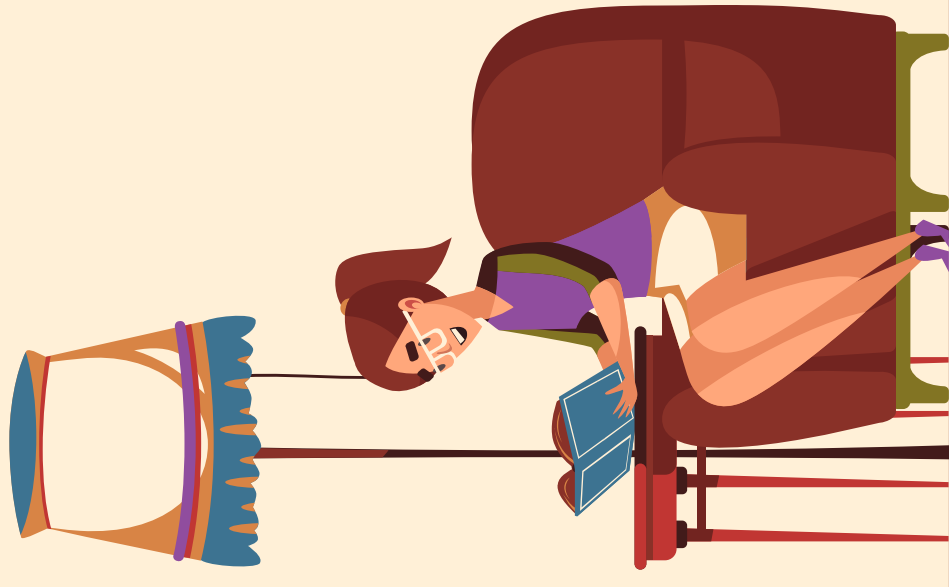
Menu Driven

Program:

1. Add Books
2. Display Books
3. Search Books
4. Issue Books
5. Return Books
6. Exit

Logic:

- User enters choice.
- Program executes corresponding block
- Returns to menu
- Stops only when choice = 6



Functional Modules

01

Add Book

Takes ID, Name, Quantity
Stores in Arrays
Increments count

02

Display Books

Loops from 0 to count -1
Displays all book details

03

Search Book

User enters ID
Loop checks matching ID
Displays book if found

04

Issue Book

Checks if quantity > 0
Decreases Quantity by 1

05

Return Book

Increases quantity by 1

06

Exit

Exits system



DEMONSTRATION



Working Code

```

1 #include <stdio.h>
2 int main()
3 {
4     int book_id[10];
5     char book_name[10][50];
6     int book_quantity[10];
7     int choice;
8     int count = 0;
9
10    while(1)
11    {
12        printf("\n-----LIBRARY MANAGEMENT SYSTEM-----\n\n");
13        printf("1. Add Book\n");
14        printf("2. Display Books\n");
15        printf("3. Search Books\n");
16        printf("4. Issue Book\n");
17        printf("5. Return Book\n");
18        printf("6. Exit\n\n");
19        printf("Enter your choice: ");
20        scanf("%d", &choice);
21
22        // add books
23        if (choice == 1)
24        {
25            printf("\nAdding a Book\n-----\n\n");
26            printf("Enter BookID : ");
27            scanf("%d", &book_id[count]);
28
29            printf("Enter Book name : ");
30            scanf(" %[\n]", book_name[count]);
31
32            printf("Enter qty: ");
33            scanf("%d", &book_quantity[count]);
34
35

```

```

}
// display books
if (choice== 2)
{
    printf("\n%-5s %-20s %-5s\n", "ID", "Name", "Quantity");
    printf("-----\n\n");
    for (int i = 0; i < count; i++)
    {
        printf("%-5d %-20s %-5d\n",
            book_id[i], book_name[i], book_quantity[i]);
    }
}

// search book
if (choice==3)
{
    int ids;
    printf("\nEnter the ID of the Book: ");
    scanf("%d", &ids);
    for (int i = 0; i < count; i++)
    {
        if (book_id[i]==ids)
        {
            printf("\nBOOK FOUND! \n\n");
            printf("ID: %d\n", book_id[i]);
            printf("Name: %s\n", book_name[i]);
            printf("Qty: %d\n", book_quantity[i]);
        }
    }
}

// Issue the book
if (choice==4)
{
    int id;
    printf("\nEnter the book id you want to issue : ");
    scanf("%d", &id);
    for (int i = 0; i < count; i++)
    {

```

```

        if (book_id[i]==id)
        {
            if (book_quantity[i]>0)
            {
                book_quantity[i]--;
                printf("Book issued \n\n");
                break;
            }
            else
            {
                printf("Enter Valid BookID!\n\n");
            }
        }
    }

    // Return book
    if (choice==5)
    {
        int id;
        printf("\nEnter the Book id : ");
        scanf("%d", &id);
        for (int i = 0; i < count; i++)
        {
            if (book_id[i]==id)
            {
                book_quantity[i]++;
                printf("Book Returned \n\n");
            }
        }
    }

    // Exit
    if(choice == 6)
    {
        break;
    }
}

```

OUTPUT

ADDITION OF BOOK

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display Books
3. Search Books
4. Issue Book
5. Return Book
6. Exit
Enter your choice: 1

Add a book to the shelf:-
Enter book id :- 1234
Enter book name :- Rohit
Enter quantity:- 4

Book added successfully!!

Number of Rohit books available now are :- 4
```

DISPLAY BOOKS

```
===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Display Books
3. Search Books
4. Issue Book
5. Return Book
6. Exit
Enter your choice: 2

ID      Name      Quantity
-----
1234    Rohit         4
```

SEARCH BOOK

```
Enter the ID of the Book: 1234

BOOK FOUND!

ID: 1234
Name: Rohit
Qty: 4
```


OUTPUT

ISSUE BOOK

```
===== LIBRARY MANAGEMENT SYSTEM =====  
1. Add Book  
2. Display Books  
3. Search Books  
4. Issue Book  
5. Return Book  
6. Exit  
Enter your choice: 4  
  
Enter the book id you want to issue : 1234  
Book issued !!
```

RETURN BOOK

```
===== LIBRARY MANAGEMENT SYSTEM =====  
1. Add Book  
2. Display Books  
3. Search Books  
4. Issue Book  
5. Return Book  
6. Exit  
Enter your choice: 5  
  
Enter the Book id : 1234  
Book Returned !!
```

EXIT

```
===== LIBRARY MANAGEMENT SYSTEM =====  
1. Add Book  
2. Display Books  
3. Search Books  
4. Issue Book  
5. Return Book  
6. Exit  
Enter your choice: 6  
  
=== Code Execution Successful ===
```

Thank
You

